LANI™
POWER FROM ABOVE

Tennessee TECH

PROJECT HELIOS / TEAM 6

Retrofit Kit Installation Manual

2020 Club Car Precedent electric golf car

Remove the existing roof and battery system. Install the LANI-EV solar roof, route the wiring, and mount the replacement battery and AC-to-DC battery charger.

Technical review draft

Electrical connections, cutting locations, and several hardware specifications still need confirmation.

Prepared by	Tennessee Technological University, ECE Department
	Capstone Design, Project HELIOS
Project partner	LANI-EV
Version	Revision 1.3
Issue date	November 12, 2026

Contents

About this manual 2

Tools and parts to prepare 3

Prepare the vehicle for work 4

Disconnect and remove the existing batteries 5

Remove the existing roof 6

Route the solar wiring 7

Install the solar mounting frame 9

Install the gasket and solar panel 11

Install the battery, charging port, and charger 13

Electrical connections and first power-up 15

Installation checks and handover 17

About this manual

This manual explains how to install the LANI-EV retrofit kit on the electric golf car identified on the cover. It covers roof removal, solar-roof mounting, cable routing, and battery installation. It is written for an installer who can identify vehicle electrical components and use the required tools safely.

Read the full procedure before starting. Use the vehicle service manual and the instructions supplied with the battery, solar panel, and AC-to-DC battery charger to confirm electrical ratings, connection order, and tightening torques. If these instructions disagree, resolve the difference with LANI-EV before continuing.

What this version covers

1. Prepare the vehicle and isolate the existing battery system.
2. Remove the original canopy, meaning the vehicle's roof.
3. Route the solar wiring through the rear roof supports.
4. Install the mounting frame, rubber gasket, solar panel, and covers.
5. Prepare the battery compartment and mount the battery, charging port, and electronics.
6. Connect the kit using the separate wiring diagram, complete the installation checks, and follow the approved first-power-up procedure.

Words used in this manual

Term	Meaning
Canopy support	The upright structure that holds the roof.
Grommet	A protective insert fitted into a drilled hole to keep the hole edge from damaging a cable.
Gasket	The rubber strip between the solar panel and mounting frame.
Polarity	The positive (+) and negative (–) identity of an electrical connection.
Torque	The specified turning force used to tighten a fastener.
TOW/RUN switch	The vehicle switch labeled TOW and RUN. Its use must follow the vehicle service manual.
AC-to-DC battery charger	Converts alternating-current (AC) power from an outlet into direct-current (DC) power to charge the battery. Also called the converter.

How to use the confirmation notes

Items C01–C10 identify missing information. Each note explains what must be confirmed and where it affects the work. The complete list is at the end of this manual. Resolve each note before performing the affected step. Numbered lists give the installation actions; confirmation notes identify details needed to carry them out.

Tools and parts to prepare

Check the tools against the actual vehicle and kit before removing parts. Keep removed hardware labeled so its original location is clear.

Tool or material	Use or detail
Phillips & Flathead screwdrivers	Various sizes.
Two adjustable wrenches	For canopy hardware. Use correctly sized combination wrenches where appropriate.
Socket wrench and sockets	Canopy and mount hardware. Sizes: .
Insulated 5/8-inch socket and ratchet	Listed for the battery hold-down nuts. Terminal sizes depend on the battery.
Torque wrench	Tighten fasteners to confirmed values.
Drill and suitable bits	Wire passages and mounting holes. Sizes: .
Fish wire or pull string	Pull the solar wiring through the rear supports.
Cable grommets	Protect wiring where it passes through drilled openings.
Scissors or utility knife	Cut the rubber gasket.
Safety glasses and insulated tools	Use for battery and electrical work. Select any additional protective equipment for the specific battery and task.
Shop vacuum	Keep the laboratory clean and prevent debris build-up in the cart.
Camera	Document procedure, vehicle state, and progress.
Wheel chocks	Keep the vehicle from rolling during work.

Kit parts to prepare

Solar mounting frame; front spacers; supplied mounting bolts; rubber gasket; solar panel; front and rear cover pieces; cover screws; wiring; replacement battery and brackets; start button and battery display; charging port; metal mounting plate; and AC-to-DC battery charger.

This is a preparation list, not a verified packing list. Check quantities, part numbers, ratings, and supplied hardware against the actual kit.

Prepare the vehicle for work

Warning. Improper battery work can cause serious injury, burns, electrical faults, or equipment damage.

Do not allow a tool to bridge battery terminals or exposed connections.

1. Turn the key switch OFF and remove the key.
2. Place the Forward/Reverse control in Neutral.
3. Set the parking brake and chock the wheels.
4. Follow the vehicle and charger instructions to disconnect external charging power and prepare the system for service.
5. Place the TOW/RUN switch in TOW before disconnecting or connecting batteries. Confirm this setting against the vehicle service manual.
6. Remove rings, watches, chains, and loose clothing that could contact electrical parts or catch on components.
7. Use insulated tools near batteries and electrical connections. Wear safety glasses and the protective equipment specified for the task.

Warning. The TOW setting is not proof that every electrical connection is de-energized. Use the manufacturer's isolation procedure for the vehicle, battery, charger, and solar equipment before handling conductors.

Before drilling or cutting

Inspect both sides of the proposed opening. Identify nearby wiring, hardware, and structural parts. Move or protect anything that the tool could damage. Do not drill or cut until the location and dimensions are confirmed.

Handling the roof and panel

Use two people for the mounting-frame installation. Arrange suitable assistance for lifting and supporting the canopy, panel, and battery. Prepare a clear resting area before removing large parts.

Confirm before proceeding (C01 and C03). Confirm the exact vehicle model, isolation procedure, discharge delay, and battery removal method.

Disconnect and remove the existing batteries

Tools and materials: Insulated tools; 5/8-inch socket and ratchet for the listed hold-down nuts; correctly sized terminal tools; torque wrench for reassembly.

Complete Prepare the vehicle for work before starting this section.

1. Identify and label the existing battery cables. Record their connection locations before removal.
2. Confirm that the TOW/RUN switch is in TOW and the required isolation steps are complete.
3. Disconnect the negative (–) cable first, following the vehicle service procedure. Keep the disconnected cable from contacting a terminal.
4. Observe the manufacturer's discharge waiting period before continuing. The provisional delay is **90 seconds**; confirm it for the installed controller before use. Time alone does not prove the system is safe to touch.
5. Follow the vehicle service procedure for the remaining cable disconnections and checks. Confirm the complete sequence before starting (C03).
6. Remove the battery hold-down nuts using the listed 5/8-inch socket if it fits the installed hardware.
7. Remove the batteries using the manufacturer-approved handling method. Keep the compartment clear for inspection and installation.

If the original batteries are already missing

Record that condition. Inspect the remaining cables and compartment before proceeding. Do not treat loose or unidentified cables as safe simply because no battery is visible.

Before moving on

Check that the existing batteries and required restraints are removed, disconnected conductors are protected, and no unidentified electrical source remains connected to the work area.

Remove the existing roof

Tools and materials: Socket wrench for the rear support bolts; 8mm for the mounts and 10mm for the base.

1. Support the canopy so it cannot drop or shift when the fasteners are removed.
2. At each front canopy support, remove the two nuts and two bolts. Keep the hardware labeled.
3. Remove the nuts and bolts that attach the canopy to the rear supports.
4. With suitable assistance, lift the canopy clear of the supports. Set it on a stable, flat surface away from the work area.



Figure 1: Vehicle with the original roof removed.

Before moving on

Inspect the exposed roof supports and mounting points. Resolve damage or missing hardware before fitting the replacement mount.

Route the solar wiring

Tools and materials: Drill; bit matched to the selected grommet; fish wire or pull string; grommets.

Keep the solar circuit isolated according to the equipment instructions while handling its connectors.

1. Remove the rear roof holding brackets shown below.
2. Separate the solar wiring connectors at the mount, as shown in the reference photos. Label the matching connections so the wiring can be routed separately without losing its identification.

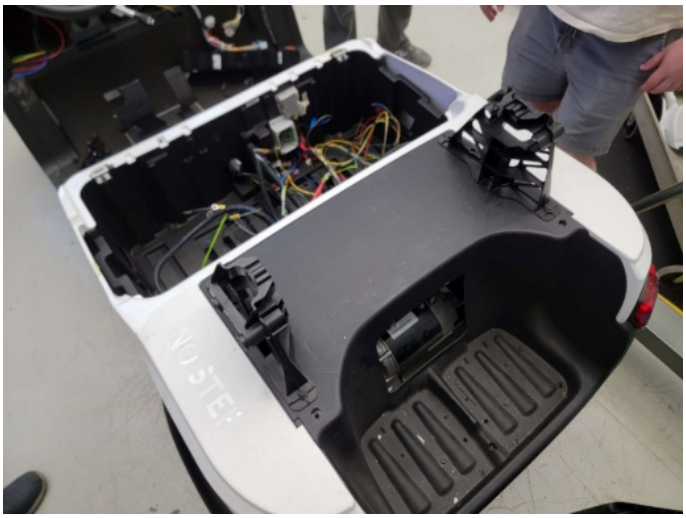


Figure 2: Rear roof holding brackets removed for wiring access.

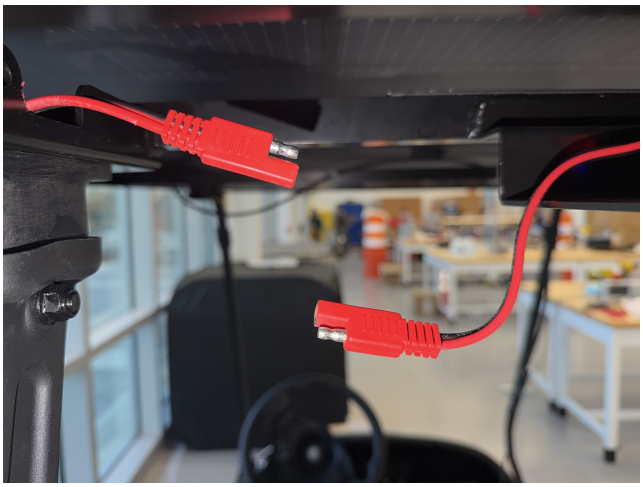


Figure 3: Solar wiring connectors separated for routing.



Figure 4: Reference view of wiring at a roof support.

Pull the wires through the rear supports

3. Feed a pull string or fish wire through each rear canopy support from top to bottom.
4. Attach the solar wiring to the pull string. Pull it gently through the support without damaging the insulation or connectors.
5. Route the wires from the bottom of each support toward the corresponding rear wheel well. Reattach the holding brackets without pinching the wires.
6. Choose an opening location in each rear wheel well that leads into the battery compartment. Inspect behind both locations before drilling.
7. Drill the confirmed opening in each wheel well. Fit the selected protective grommet, then route the cable through it into the battery compartment.

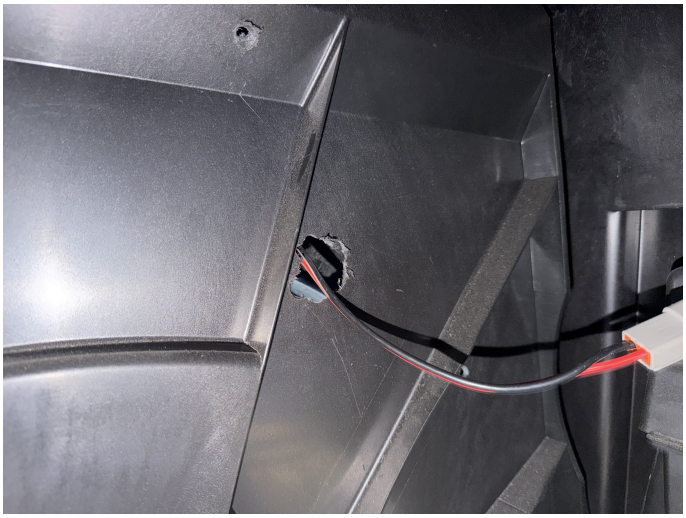


Figure 5: Wheel-well opening location.

Warning. The provisional opening range is **1–1.5 inches**. This is not a final drill size. Confirm the location, cable clearance, and grommet's required hole diameter before drilling either wheel well.

Before moving on

Check both cable paths for sharp edges, pinching, tire clearance, and strain at the connectors. Keep the cable ends identified and isolated until the approved electrical connection procedure is available.

Install the solar mounting frame

Tools and materials: Two installers; supplied mounting hardware; correctly sized wrenches or sockets.

Use the hardware supplied for this kit. Do not substitute bolts without confirming their size and suitability with LANI-EV.

1. Place the front spacers over the front canopy support mounting holes on the driver and passenger sides.
2. Position the solar mounting frame on the front and rear supports. Check that all holes line up before inserting fasteners.
3. Insert the supplied front bolts through the frame, spacers, and support holes. Leave them hand-tight.
4. Insert the supplied rear bolts through the frame and rear support holes. Leave them hand-tight.
5. Check the position of the entire frame. Tighten the front fasteners, then the rear fasteners, to the confirmed kit torque values.
6. Inspect each mounting point. Confirm that the frame is secure and sits correctly before installing the panel.

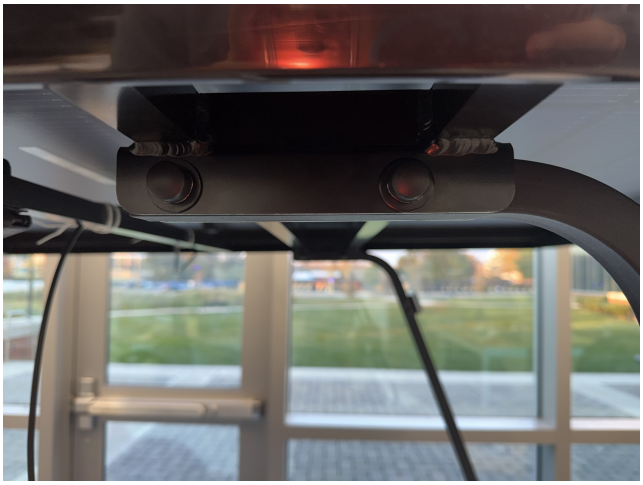


Figure 6: Front attachment reference.



Figure 7: Rear attachment reference.

Inspect the mounted frame



Figure 8: Solar mounting frame installed on the vehicle.

- All specified fasteners are installed and tightened to confirmed values.
- Spacers sit in their intended locations without gaps caused by trapped wiring.
- Wiring is not pinched at the supports or frame.
- The frame is aligned and ready to receive the gasket and panel.

Install the gasket and solar panel

Tools and materials: Allen key ($\frac{5}{32}$") and wrench to tighten the bolt, correctly sized screwdriver; scissors or utility knife; supplied gasket and cover hardware.

Prepare the mount and gasket

1. Remove the four cover screws from both the front and rear of the mount. Keep them for reinstallation.
2. Cut the rubber gasket into two pieces as specified for the kit. Confirm that two equal halves provide the required lengths ($\approx 39.5\,\mathrm{in.}$) before cutting.
3. Place the gasket pieces at the front and rear panel contact surfaces. Keep them flat, without bunching or gaps.
4. Place the rubber gasket over the existing frame before the panel is attached.

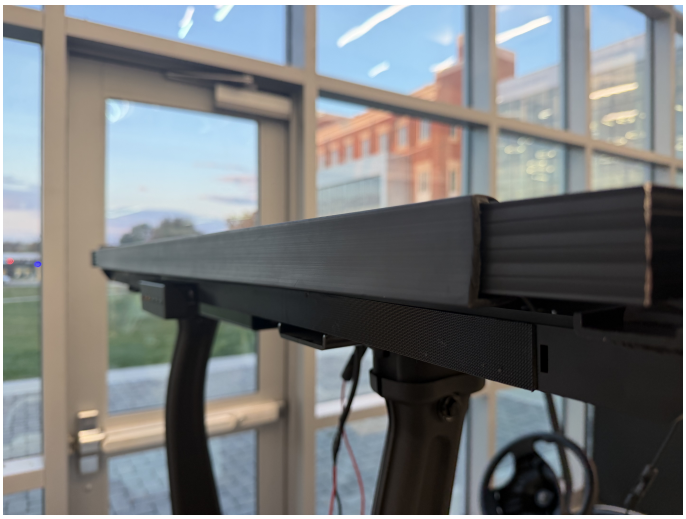


Figure 9: Rubber gasket at the panel contact surface.

Set the panel on the mount

4. Orient the panel so its positive (+) and negative (–) leads exit toward the rear of the vehicle.
5. With suitable assistance, lower the panel onto the mount. Make sure it rests on the intended front and rear gasket surfaces.

Secure the panel

6. Place the front cover over the panel's front edge.
7. Place the rear cover over the panel's rear edge.
8. Reinstall the saved screws. Tighten evenly, alternating sides, using the confirmed kit specification. Check that the panel is retained without excessive gasket compression.

Warning. Do not overtighten the cover screws. Excessive clamping can damage the panel or frame. Confirm the required tightening value rather than relying only on how firmly the panel feels held.

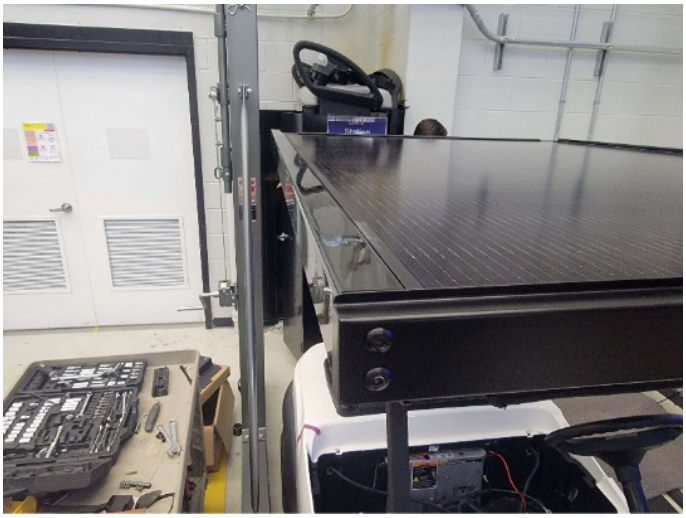


Figure 10: Panel cover installed at the edge of the solar panel.

Identify the panel connections

9. Identify and label the panel's positive and negative leads and their matching connections at the rear mount using the separate kit wiring diagram.
10. Keep the connectors protected and isolated until the electrical connection stage. Do not identify polarity from wire color or photos alone.

Warning. Keep these connections isolated until the manufacturer-approved connection order is confirmed.

Install the battery, charging port, and charger

Clean and inspect the compartment

1. Inspect the compartment for loose debris, damage, and evidence of battery leakage.
2. Remove ordinary dry debris using an appropriate method. If battery residue or liquid is present, follow the battery manufacturer's spill-cleanup procedure before continuing. Do not treat suspected acid residue as ordinary debris to vacuum.
3. Identify the raised plastic features that prevent the replacement battery from sitting correctly. Obtain the approved trimming locations, limits, and tool choice before removing material.
4. Trim only the approved plastic features using the specified tool and cutting limits. Remove debris and inspect the mounting surface. Check that the battery can sit level and that no structure or hidden wiring has been damaged.

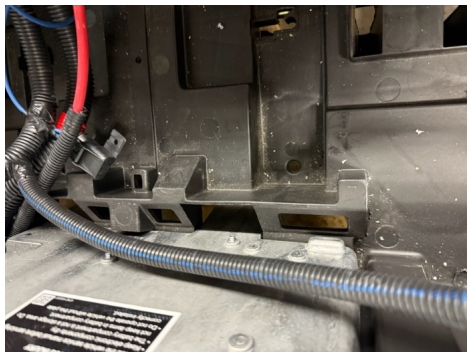


Figure 11: Battery compartment before trimming.

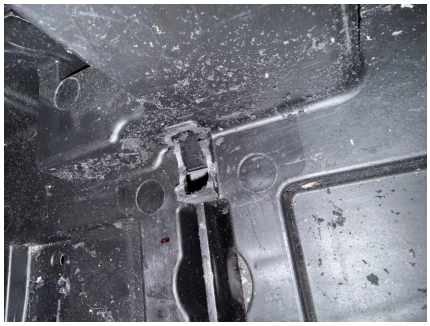


Figure 12: Detail of trimmed plastic.



Figure 13: Second view of the trimmed compartment.

Route the start button and display wiring

1. Route the start-button and battery-display wiring from the console toward the battery compartment.
2. Feed the end with the exposed conductors through the confirmed opening beneath the plastic body. Keep the conductors protected and unconnected.
3. Route the cables along a protected path under the floor mat. Confirm that the mat, feet, seat, and body panels cannot crush or abrade the cables; the mat alone is not protection against crushing.

Replace the charging port

1. Identify the existing charging port and its wiring while the system is isolated.
2. Follow the approved kit procedure to disconnect or retain the old port and wiring. Do not leave a loose or potentially live port tucked inside the compartment.
3. Position the new charging port on the driver's side. The battery is positioned on the passenger's side.
4. Feed the new port's cable through the existing port opening. Confirm the drilling pattern and clearance behind the panel, then drill the mounting holes and secure the port with the specified hardware.

Mount the battery and AC-to-DC charger

1. Position the battery with its mounting brackets. Mark the approved mounting-hole locations.
2. Position the metal plate beside the battery. Leave the clearance required by the charger instructions for cooling and cable routing.
3. Confirm the hole locations, clearances, fasteners, and retention method before drilling. Install the battery hardware and secure the plate to the floor.
4. Mount the AC-to-DC battery charger on the plate using its specified orientation, spacing, and hardware.

Electrical connections and first power-up

The AC-to-DC battery charger (converter) charges the battery from an electrical outlet. The solar charging circuit is a separate connection path; follow the kit wiring diagram for its components and connections.

Before making connections

1. Complete the battery, charger, charging-port, and solar-panel mounting steps. Keep outlet power disconnected and the battery and solar circuits isolated using the equipment instructions.
2. Obtain the approved *LANI-EV Retrofit Kit Wiring Diagram*, maintained as a separate document. Check that it matches the installed kit and record its document number and revision in the installation record.
3. Match the battery, charger, solar equipment, and charging-port model numbers to the diagram. Identify each cable and connector by its diagram designation.
4. Read the component instructions and the diagram's connection sequence before attaching any conductors. Resolve missing connector assignments, protection requirements, or tightening values before continuing.

Connect the kit wiring

5. Route and label the charging-inlet cable to the charger's AC input, and the charger output cable to the battery charging connection identified in the diagram. Keep AC input wiring distinct from DC battery wiring.
6. Route and label the start-button and battery-display cables to their designated battery control connectors. Route the vehicle power cables and solar cables to the destinations identified in the diagram.
7. Following the approved connection sequence, attach each labeled cable to its designated connector or terminal. Check polarity and connector identification before each connection; tighten terminal hardware to the specified value and record it.
8. Fit the specified cable restraints, protective covers, and strain relief. Check that no cable is pinched or pulled tight and that no exposed conductor can contact the frame or another terminal.
9. Compare every completed connection with the diagram. Record the inspection results before proceeding to first power-up.

First power-up — procedure to complete

1. Complete the mechanical and cable checks in Installation checks and handover. Record the wiring-diagram revision and reviewer.

2. Perform the pre-power electrical checks specified for the installed equipment. Record each required measurement and compare it with its approved limit. **TBD: test points, checks, and acceptance limits.**

3. Apply power using the approved sequence for the battery, vehicle, solar circuit, and outlet charger. **TBD: exact switch positions, connection order, delays, and expected indicators.**

4. Verify battery-display and start-button operation, outlet charging, solar charging, and vehicle operation using the approved test instructions. Record the results. **TBD: test conditions and pass/fail criteria for each function.**

5. If a check fails or a fault appears, stop the test and follow the equipment's shutdown instructions. Record the fault and resolve it before repeating the test. **TBD: kit shutdown sequence and fault response.**

6. After all checks pass, complete the installation record and document any operating restrictions before handing over the vehicle.

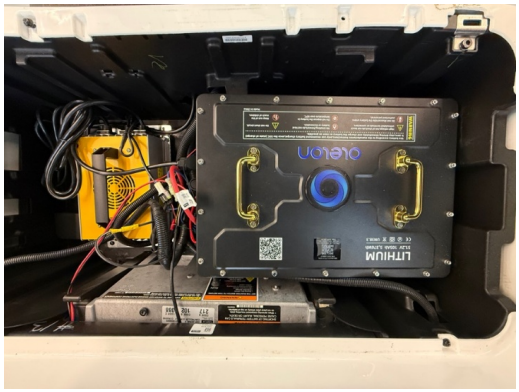


Figure 14: Battery, AC-to-DC charger, and adjacent electronics. Use the separate wiring diagram for connection details.

Installation checks and handover

Complete the following checks after installation. Record the results, then complete the approved first-power-up procedure before releasing the vehicle.

Mechanical and cable checks

1. Inspect the roof supports, spacers, frame, and panel covers for correct installation.
2. Check each fastener against its specified torque and record the result.
3. Check that the panel is seated on the gasket and retained without damage.
4. Inspect wire passages for suitable edge protection and confirm that cables are clear of pinch points, sharp edges, tires, and moving parts.
5. Check that the battery, brackets, mounting plate, charger, and charging port are secured using approved hardware.
6. Remove cutting debris, loose fasteners, and tools from the compartment.

Electrical work that must be completed before release

1. Record the battery, panel, AC-to-DC charger and solar charging equipment, and charging-port model numbers.
2. Record the separate wiring diagram document number and revision, and verify every connection against it.
3. Complete the specified isolation, polarity, protection, and pre-power checks.
4. Perform the approved start-up and functional checks. Record results and any faults.
5. Close every confirmation item that applies to the installed kit.

Installation record

Vehicle model and serial number	
Kit version or serial number	
Battery model	
Solar panel model	
Charger and solar equipment models	
Wiring diagram number/revision	
Installer and installation date	
Reviewer and review date	
Unresolved items or restrictions	